



Fundamentals of Optics

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Rules

- Final exam: 30%
- Mid-term exam: 30%
- Homework: 25%
- Attendance: 15%

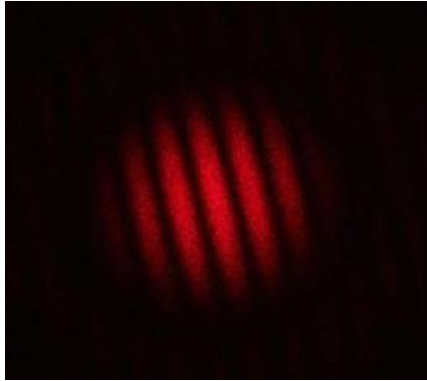
Reference books

Optics, by Eugene Hecht
光学, 赵凯华、钟锡华

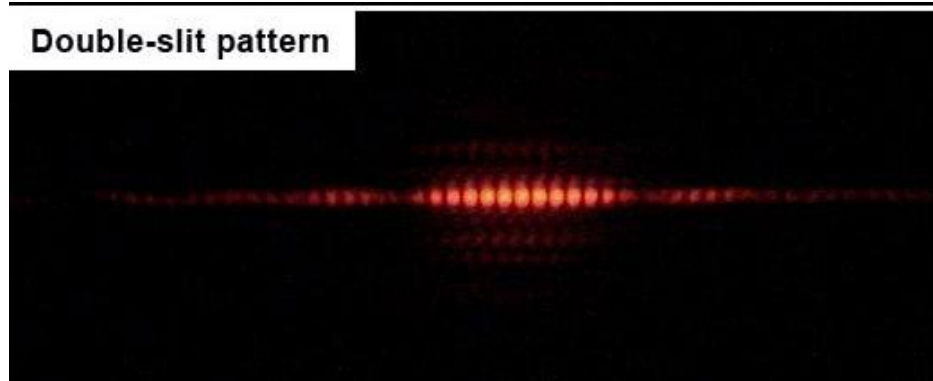
CHAPTER 6

THE SUPERPOSITION OF WAVES

Study the phenomena of **interference**, **diffraction** and **polarization**



Interference of light through two narrow holes



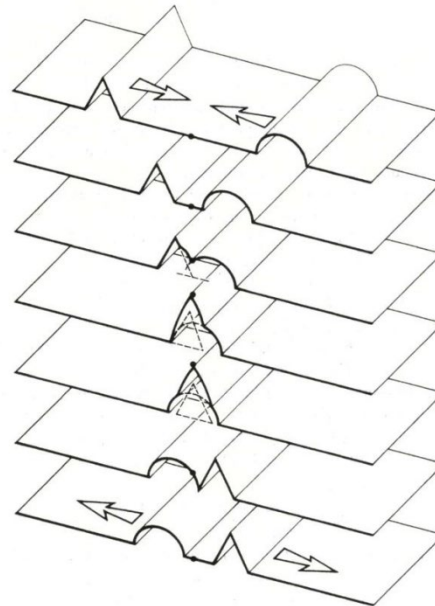
Multi-Slit Diffraction Experiments



Rainbow effect from cross polarization of clear plastic objects.

Share a common conceptual basis: **lightwave overlap**

The resultant disturbance at any point in a medium is the algebraic sum of the separate constituent waves in **linear** systems.



Content

- ❑ The Addition of Waves of the Same Frequency
- ❑ The Addition of Waves of Different Frequency
- ❑ A harmonic Periodic Waves
- ❑ Nonperiodic Waves

The addition of Waves of the Same Frequency

The algebraic method

An harmonic disturbance $E(x, t) = E_0 \sin[\omega t - (kx + \epsilon)]$

E_0 : the amplitude of the harmonic disturbance propagating along the positive x-direction.

Separate the space and time parts of the phase

$$E(x, t) = E_0 \sin(\omega t + \alpha) \quad \text{and} \quad \alpha(x, \epsilon) = -(kx + \epsilon)$$

Suppose that there are two waves with the same frequency and speed:

$$E_1 = E_{01} \sin(\omega t + \alpha_1) \quad \text{and} \quad E_2 = E_{02} \sin(\omega t + \alpha_2)$$

$$\alpha_1(x_1, \epsilon_1) = -(kx_1 + \epsilon_1) \quad \alpha_2(x_2, \epsilon_2) = -(kx_2 + \epsilon_2)$$

The result disturbance is the linear superposition of these waves:

$$E = E_1 + E_2$$

We can anticipate a sinusoidal function of frequency ω with an amplitude (E_0) and phase (α).

$$\begin{aligned}
 E &= E_1 + E_2 = E_{01}\sin(\omega t + \alpha_1) + E_{02}\sin(\omega t + \alpha_2) \\
 &= E_{01} (\sin\omega t \cos \alpha_1 + \cos \omega t \sin \alpha_1) \\
 &\quad + E_{02}(\sin\omega t \cos \alpha_2 + \cos \omega t \sin \alpha_2) \\
 &= (E_{01}\cos \alpha_1 + E_{02}\cos \alpha_2)\sin\omega t \\
 &\quad + (E_{01}\sin \alpha_1 + E_{02}\sin \alpha_2) \cos \omega t
 \end{aligned}$$

Let $E_0 \cos \alpha = E_{01} \cos \alpha_1 + E_{02} \cos \alpha_2$

$$E_0 \sin \alpha = E_{01} \sin \alpha_1 + E_{02} \sin \alpha_2$$

$$E_0^2 = E_{01}^2 + E_{02}^2 + 2E_{01}E_{02} \cos(\alpha_2 - \alpha_1)$$

$$\tan \alpha = \frac{E_{01} \sin \alpha_1 + E_{02} \sin \alpha_2}{E_{01} \cos \alpha_1 + E_{02} \cos \alpha_2}$$

The total disturbance:

$$E = E_0 \cos \alpha \sin \omega t + E_0 \sin \alpha \cos \omega t$$

$$E = E_0 \sin(\omega t + \alpha)$$

The composite wave is harmonic and of the same frequency as the constituent with different amplitude and phase.

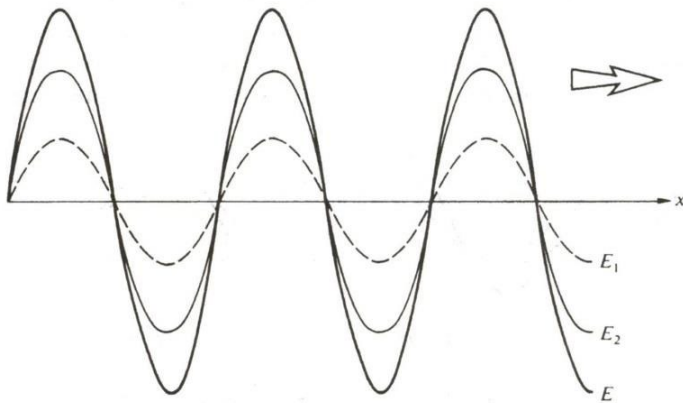
$$E_0^2 = E_{01}^2 + E_{02}^2 + 2E_{01}E_{02}\cos(\alpha_2 - \alpha_1)$$

The flux density is proportional to its amplitude squared. But it is not simply the sum of the component of flux densities.

The interference term: $2E_{01}E_{02}\cos(\alpha_2 - \alpha_1)$

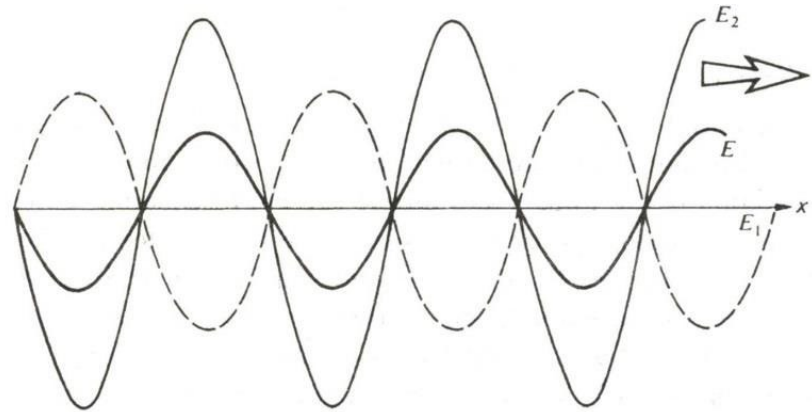
Difference in phase between the two interfering waves: $\delta \equiv \alpha_2 - \alpha_1$

In phase: $\delta = \pm 2n\pi$



Crest overlaps crest

180° out of phase: $\delta = \pm (2n+1)\pi$



Trough overlaps crest

Phase difference:

$$\delta = (kx_1 + \epsilon_1) - (kx_2 + \epsilon_2) = \frac{2\pi}{\lambda} (x_1 - x_2) + (\epsilon_1 - \epsilon_2)$$

x_1 and x_2 : the distances from the sources to the point of observation.

If the waves are initially in-phase: $\delta = \frac{2\pi}{\lambda} (x_1 - x_2)$

In the media, $n = \frac{c}{v} = \frac{\lambda_0}{\lambda}$, $\delta = \frac{2\pi}{\lambda_0} n(x_1 - x_2)$

Optical path difference: the difference in the two optical path lengths (OPL)

$$\Lambda = n(x_1 - x_2)$$

$$\delta = 2\pi \frac{\Lambda}{\lambda_0} = k_0 \Lambda$$

One route is δ radians longer than the other.

Coherent: $\epsilon_1 - \epsilon_2 = \text{constant}$ regardless of its value.

Example

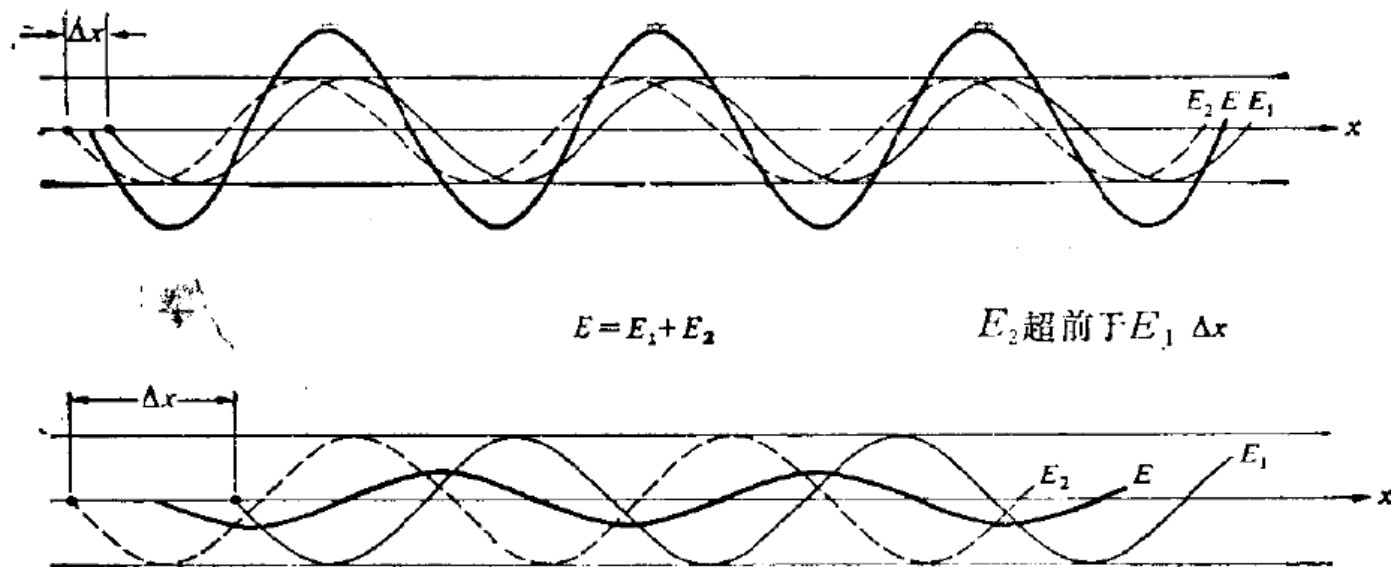
$$E_1 = E_{01} \sin[\omega t - k(x + \Delta x)] \quad E_2 = E_{02} \sin(\omega t - kx)$$

$$\epsilon_1 = \epsilon_2 \quad E_{01} = E_{02} \quad \alpha_2 - \alpha_1 = k\Delta x$$

$$E = 2E_{01} \cos\left(\frac{k\Delta x}{2}\right) \sin\left[\omega t - k\left(x + \frac{\Delta x}{2}\right)\right]$$

Case I: $\Delta x \ll \lambda$ $E_0 \approx 2E_{01}$ **Constructive interference (相长干涉)**

Case II: $\Delta x = \lambda/2$ $E_0 = 0$ **Destructive interference (相消干涉)**



The Superposition of Many Waves

Superposition of any number of coherent harmonic waves having a given frequency and traveling in the same direction leads to a harmonic wave of that same frequency.

The sum of N waves

$$E = \sum_{i=1}^n E_{0i} \cos(\alpha_i \pm \omega t)$$

is given by $E = E_0 \cos(\alpha \pm \omega t)$

where

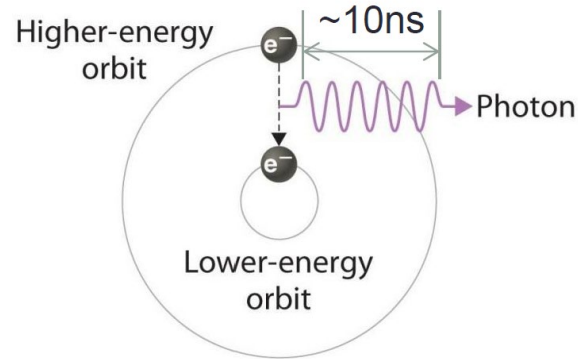
$$E_0^2 = \sum_{i=1}^N E_{0i}^2 + 2 \sum_{j>i}^N \sum_{i=1}^N E_{0i} E_{0j} \cos(\alpha_i - \alpha_j)$$

$$\tan \alpha = \frac{\sum_{i=1}^n E_{0i} \sin \alpha_i}{\sum_{i=1}^n E_{0i} \cos \alpha_i}$$

In the case of in-phase coherent sources ($\alpha_i = \alpha_j$): $E_0^2 = N^2 E_{01}^2$

Example: atomic emitters (N sources)

Imagine the photon as somehow associated with a short-duration oscillatory wave pulse.



The atoms can be thought as emitting wavetrains that have a sustained phase for up to about 10 ns, after that a new wavetrain may be emitted with a totally random phase.

Flux density:

$$I = \langle E_0^2 \rangle = \sum_{i=1}^N \langle E_{0i}^2 \rangle + 2 \sum_{j>i}^N \sum_{i=1}^N E_{0i} E_{0j} \langle \cos(\alpha_i - \alpha_j) \rangle = \sum_{i=1}^N \langle E_{0i}^2 \rangle = N E_{01}^2$$

The resultant flux density arising from N sources having random, rapidly varying phases is given by N times the flux density of any one source.

Standing Waves

$$\frac{\partial^2}{\partial x^2}(\psi_1 + \psi_2) = \frac{1}{v^2} \frac{\partial^2}{\partial t^2}(\psi_1 + \psi_2)$$

Superposition principle

The general solution of the differential wave equation:

$$\varphi(x, t) = C_1 f(x - vt) + C_2 g(x + vt)$$

Examine two harmonic waves of the same frequency propagating in opposite direction :

An incoming wave traveling to the left: $E_I = E_{0I} \sin(kx + \omega t + \epsilon_I)$

It strikes a mirror at $x=0$, the reflected wave: $E_R = E_{0R} \sin(kx - \omega t + \epsilon_R)$

The composite wave: $E = E_I + E_R$

The two waves exist simultaneously in the region between the source and the mirror.

Assume that $E_{0I} = E_{0R}$; boundary conditions: $x = 0, E = 0$; set $\epsilon_I = 0$, so $\epsilon_R = 0$.

Composite disturbance

$$E = E_{0I} [\sin(kx + \omega t) + \sin(kx - \omega t)] = 2E_{0I} \sin kx \cos \omega t$$

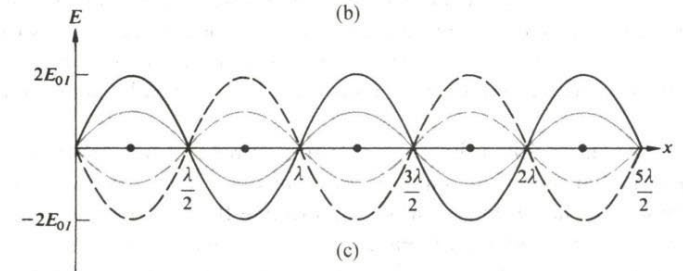
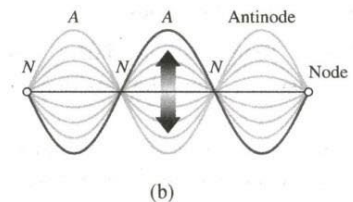
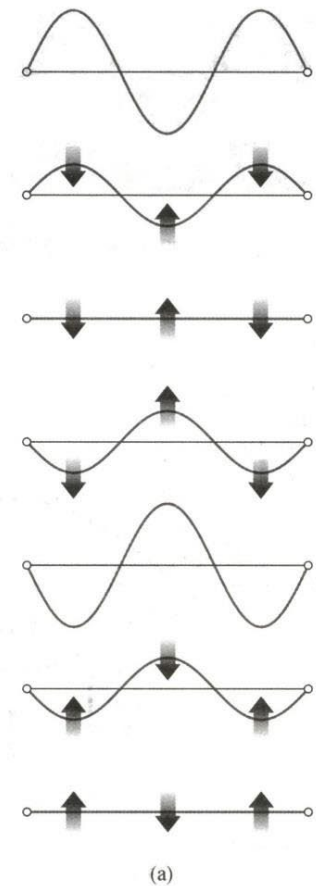
$$\sin \alpha + \sin \beta = 2 \sin \frac{1}{2}(\alpha + \beta) \sin \frac{1}{2}(\alpha - \beta)$$

Standing or stationary wave:

$$E = 2E_{0I} \sin kx \cos \omega t$$

Its profile does not move through space, because it is not of the form $f(x \pm vt)$

- ❑ At certain points (**nodes**—波节), $x=0, \lambda/2, \lambda, 3\lambda/2, \dots: E(x, t)=0$.
- ❑ Halfway between each adjacent node (**antinodes** —波腹): the amplitude has a maximum value of $\pm 2E_{0I}$.
- ❑ The disturbance $E(x, t)=0$, whenever $\cos \omega t=0$ or $t=(2m+1)\tau/4$, $m=0, 1, 2, \dots$



Standing wave water

Wiener's Experiment (Yr 1890)

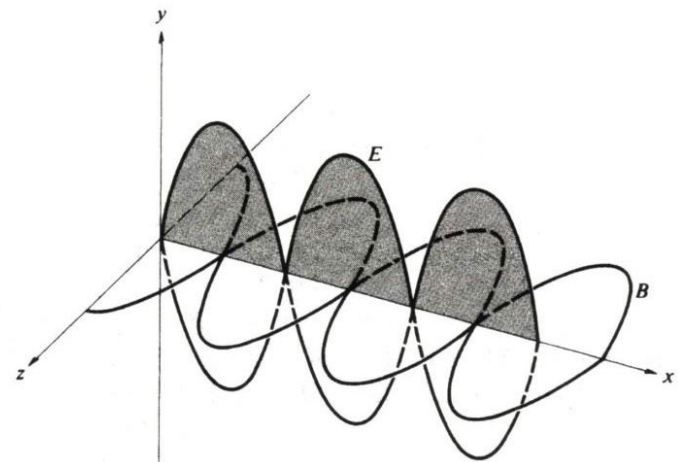
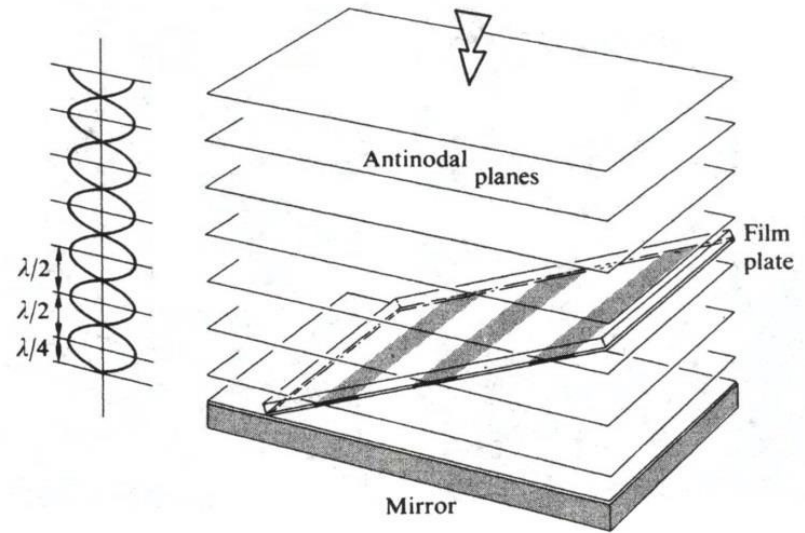
- ❑ Phenomena: the glass with emulsion is found to be blackened along a series of equidistant parallel bands.
- ❑ The black regions correspond to the place where the photographic layer had intersected the antinodal planes.

Energy conservation?

$$E(x, t) = 2E_{0I} \sin kx \cos \omega t$$

$$\begin{aligned} B(x, t) &= - \int \frac{\partial E}{\partial x} dt = -2E_0 k \cos kx \int \cos \omega t dt \\ &= -\frac{2E_0 k}{\omega} \cos kx \sin \omega t \\ &= -2B_0 \cos kx \sin \omega t \end{aligned}$$

It is the electric field that triggers the photochemical process.



The Addition of Waves of Different Frequency

Beats

$$\cos\alpha + \cos\beta = 2 \cos\frac{1}{2}(\alpha + \beta)\cos\frac{1}{2}(\alpha - \beta)$$

In the case of two waves of different frequency traveling in the same direction:

$$E_1 = E_{01}\cos(k_1x - \omega_1t) \quad \text{and} \quad E_2 = E_{01}\cos(k_2x - \omega_2t)$$

The net composite wave:

$$\begin{aligned} E &= E_{01}[\cos(k_1x - \omega_1t) + \cos(k_2x - \omega_2t)] \\ &= 2E_{01}\cos\frac{1}{2}[(k_1+k_2)x - (\omega_1 + \omega_2)t]\cos\frac{1}{2}[(k_1-k_2)x - (\omega_1 - \omega_2)t] \end{aligned}$$

Average angular frequency ($\bar{\omega}$)

$$\bar{\omega} \equiv \frac{1}{2}(\omega_1 + \omega_2)$$

Average propagation number ($\bar{k}$)

$$\bar{k} \equiv \frac{1}{2}(k_1 + k_2)$$

Modulation frequency (ω_m)

$$\omega_m \equiv \frac{1}{2}(\omega_1 - \omega_2)$$

Modulation propagation number (k_m)

$$k_m \equiv \frac{1}{2}(k_1 - k_2)$$

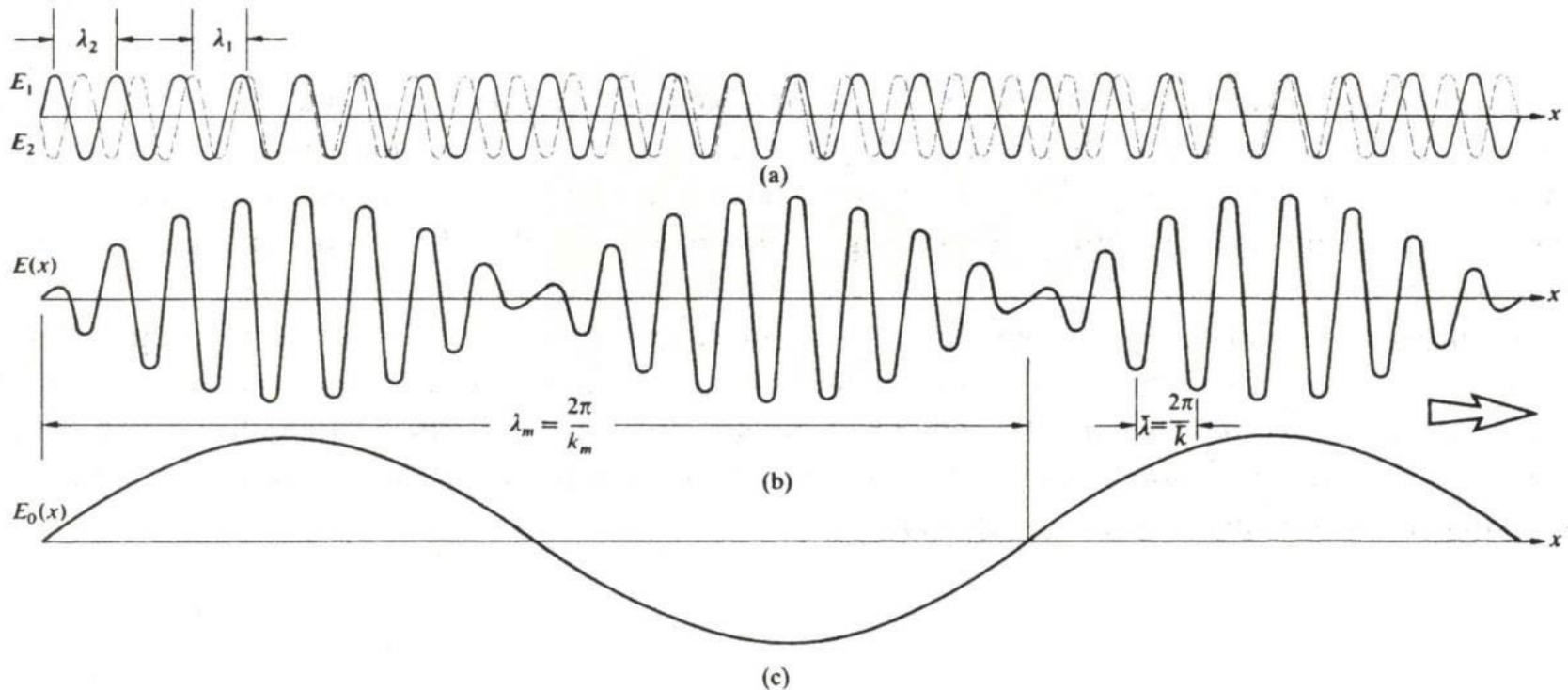
$$E(x, t) = 2E_{01} \cos(k_m x - \omega_m t) \cos(\bar{k}x - \bar{\omega}t)$$

Or rewrite as $E(x, t) = E_0(x, t) \cos(\bar{k}x - \bar{\omega}t)$

where $E_0(x, t) = 2E_{01} \cos(k_m x - \omega_m t)$

The total disturbance may be regarded as a traveling wave of frequency $\bar{\omega}$ having a time-varying or modulated amplitude $E_0(x, t)$.

For quasi-monochromatic light, $\omega_1 \approx \omega_2$, then $\bar{\omega} \gg \omega_m$ and $E_0(x, t)$ will change slowly.

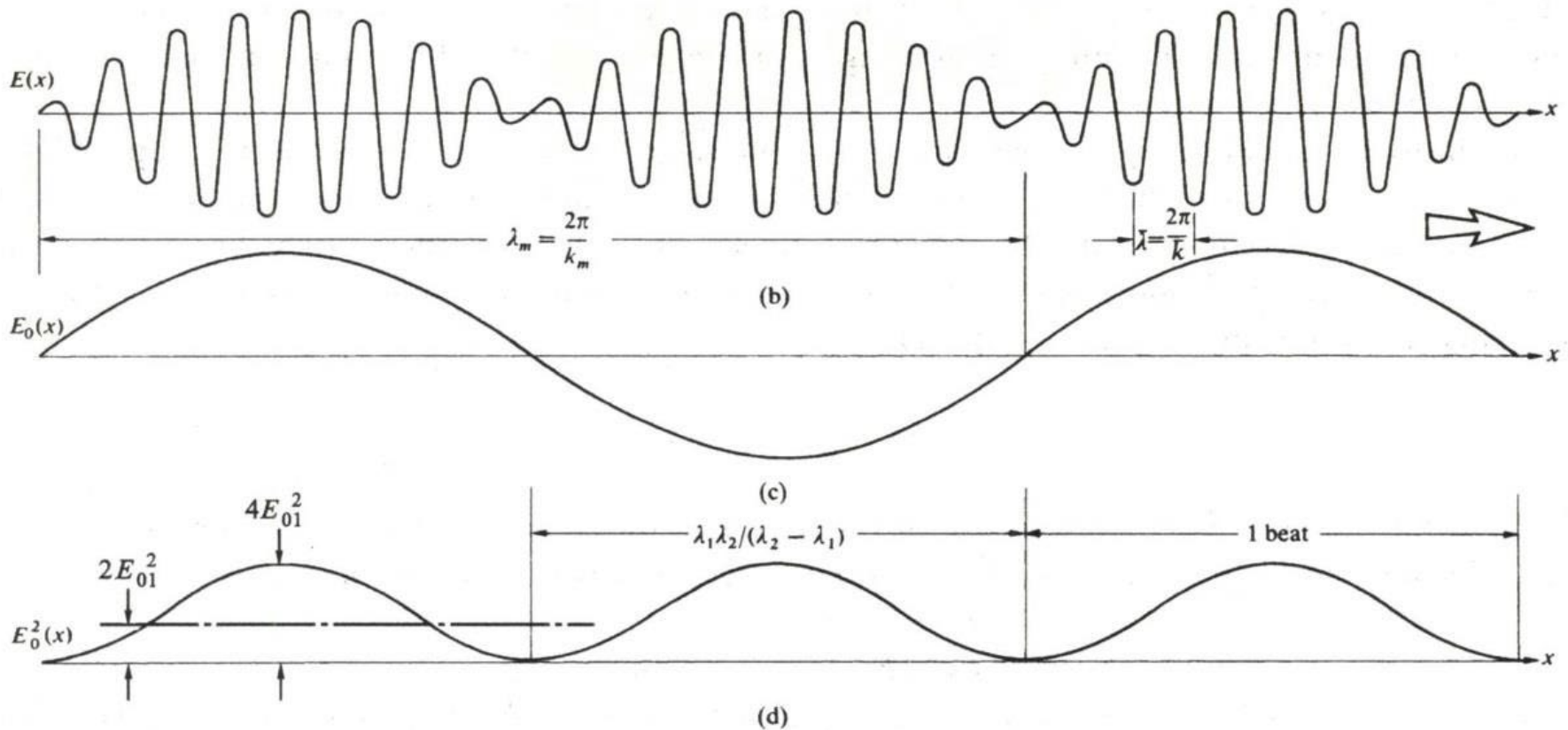


The irradiance:

$$E_0^2(x, t) = 4E_{01}^2(x, t)\cos^2(k_m x - \omega_m t)$$

$$= 2E_{01}^2(x, t)[1 + \cos(2k_m x - 2\omega_m t)]$$

Beat frequency: $2\omega_m$ or $(\omega_1 - \omega_2)$

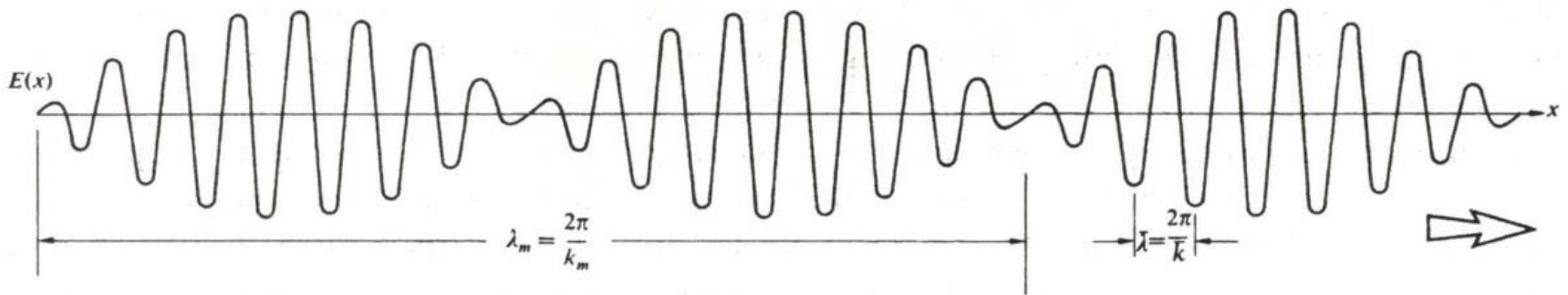


Velocity

The disturbance: $E(x, t) = E_0(x, t)\cos(\bar{k}x - \bar{\omega}t)$

It consists of a high-frequency ($\bar{\omega}$) carrier wave, amplitude-modulated by a cosine function.

Phase velocity (v) : $v = -\frac{(\partial\varphi/\partial t)_x}{(\partial\varphi/\partial x)_t} = \frac{\bar{\omega}}{\bar{k}}$ it is independent of modulation



Group velocity (v_g) : the rate at which the modulation envelope advances.

Suppose that the constituent waves, $E_1(x, t)$ and $E_2(x, t)$, advance with the same speed, $v_1 = v_2$. And the two harmonic functions having different wavelengths and frequencies. $v_g = v = v_1 = v_2$

The expression for the modulation envelope:

$$E_0(x, t) = 2E_{01}(x, t)\cos(k_mx - \omega_mt)$$

The modulation advances at a rate (**group velocity**) dependent on the phase of the envelope $(k_mx - \omega_mt)$

$$v_g = -\frac{(\partial\varphi/\partial t)_x}{(\partial\varphi/\partial x)_t} = \frac{\omega_m}{k_m} = \frac{\omega_1 - \omega_2}{k_1 - k_2} = \frac{\Delta\omega}{\Delta k}$$

In a dispersive media, ω is dependent on λ , or equivalently on k .

When frequency range $\Delta\omega$ is small, centered about $\bar{\omega}$, the **group velocity** equals to:

$$v_g = \frac{d\omega}{dk}$$

$$v_1 = \frac{\omega_1}{k_1} = \frac{c}{n(\lambda_1)} \quad v_2 = \frac{\omega_2}{k_2} = \frac{c}{n(\lambda_2)}$$

$$\begin{aligned} v_g = \frac{\Delta\omega}{\Delta k} &= \frac{v_1 k_1 - v_2 k_2}{k_1 - k_2} = \textcolor{red}{v_1} + \frac{k_2}{k_1 - k_2} (v_1 - v_2) \\ &= \textcolor{red}{v_2} + \frac{k_1}{k_1 - k_2} (v_1 - v_2) \end{aligned}$$

$$v_g = \bar{v} + \bar{k} \cdot \frac{\Delta v}{\Delta k}$$

Average velocity

$$\bar{v} = \frac{1}{2} (v_1 + v_2)$$

Average propagation number

$$\bar{k} = \frac{1}{2} (k_1 + k_2)$$

Dispersive term

$$\frac{\Delta v}{\Delta k} = \frac{v_1 - v_2}{k_1 - k_2}$$

$$v_g = \frac{d\omega}{dk} \qquad \omega = \frac{kc}{n}$$

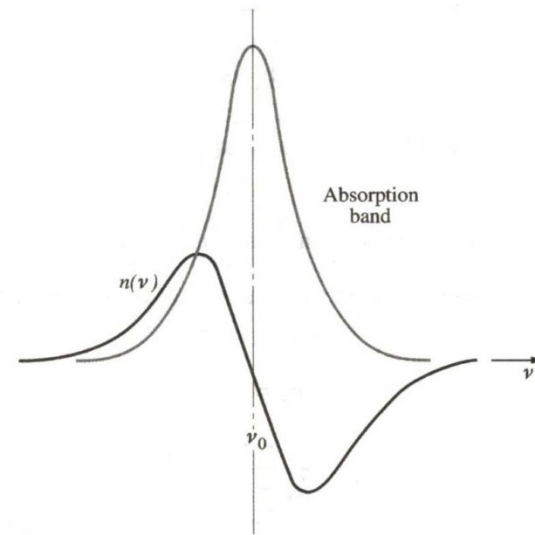
$$v_g = \frac{d\omega}{dk} = \frac{c}{n} - \frac{kc}{n^2} \frac{dn}{dk} = v \left(1 - \frac{k}{n} \frac{dn}{dk} \right)$$

Normal dispersion $v_g < \bar{v}$

No dispersion $v_g = \bar{v}$

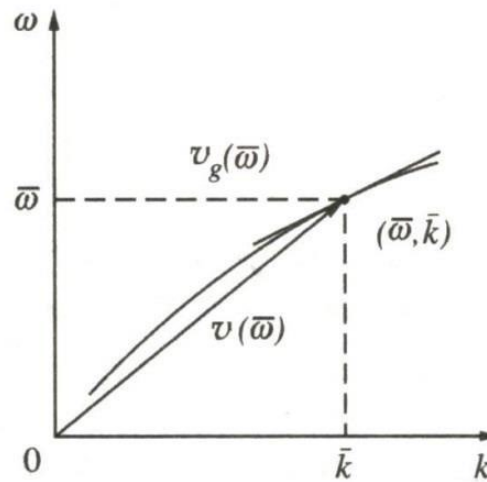
Abnormal dispersion $v_g > \bar{v}$

- $n(\nu)$ curve (typical medium)
 - ✓ Normal dispersion: $n(\nu)$ increases with increasing ν .
 - ✓ Anomalous dispersion: $n(\nu)$ decreases with increasing ν .



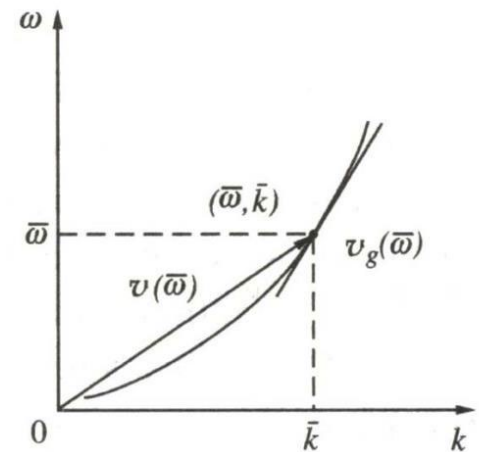
A typical representation of the frequency dependence of the index of refraction in the vicinity of an atomic resonance.

- Dispersion relation:
 - convex upward for normal dispersion
 - and concave downward for anomalous dispersion



(a)

normal dispersion



(b)

anomalous dispersion

Since $\omega = kv$

From $v_g = \frac{d\omega}{dk}$, we have $v_g = v + k \frac{dv}{dk}$

In nondispersive media: v is independent of λ .

So $v_g = v$

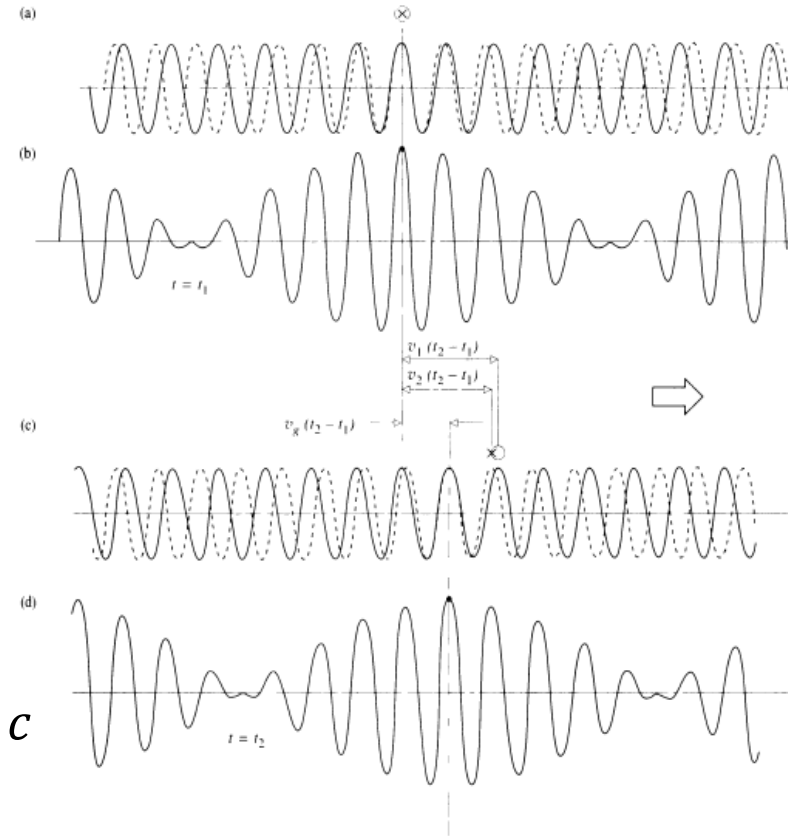
Specifically, in vacuum $\omega = kc$, $v = c$, and $v_g = c$

In dispersive media ($n(k)$): $\omega = \frac{kc}{n}$

So $v_g = \frac{c}{n} - \frac{kc}{n^2} \frac{dn}{dk}$, or $v \left(1 - \frac{k}{n} \frac{dn}{dk} \right)$

For optical media in regions of normal dispersion, $\frac{dn}{dk} > 0$, so $v_g < v$.

In regions of anomalous dispersion, $\frac{dn}{dk} < 0$, so $v_g > v$.

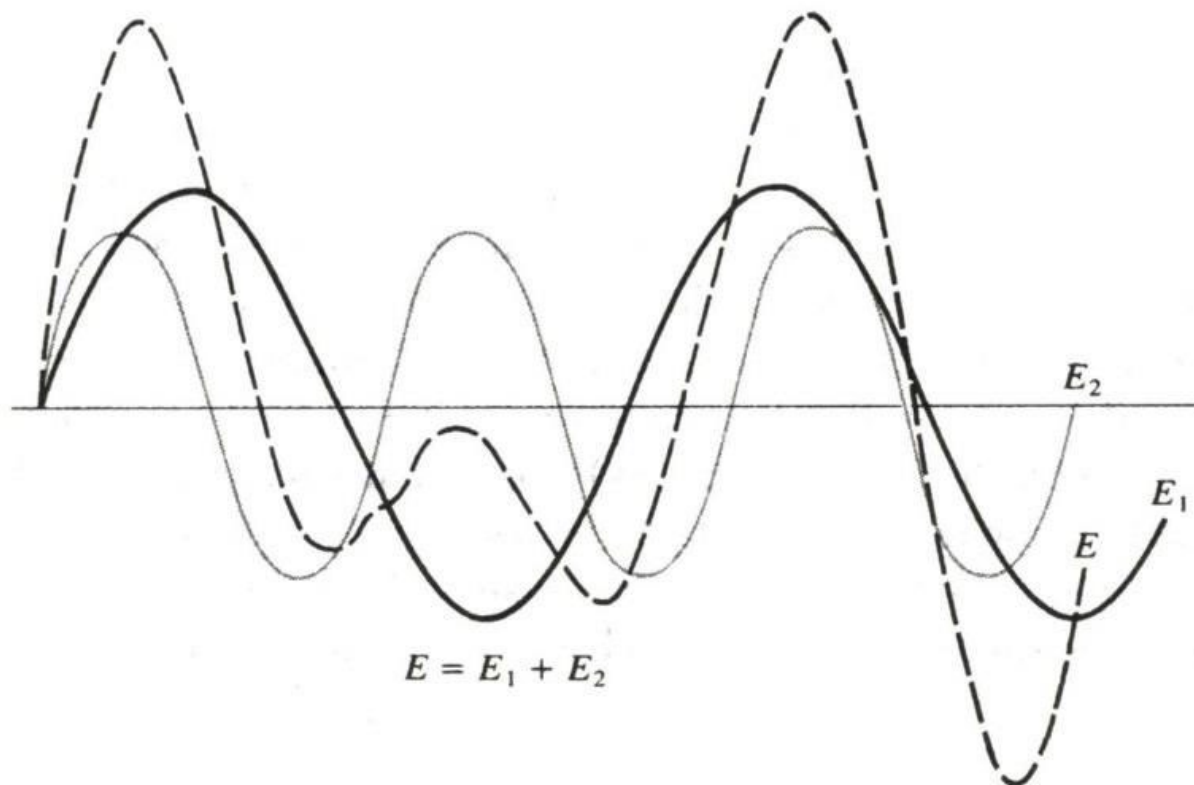


Homework

□ 7.1, 7.5, 7.19, 7.24

6.3 Anharmonic Periodic Waves-Fourier Analysis

The composite disturbance is **anharmonic** (or it is not sinusoidal), when two harmonic functions having **different amplitudes and wavelengths** superpose.



Fourier's theorem:

A function $f(x)$, having a spatial period λ , can be synthesized by a sum of harmonic functions whose wavelengths are integral submultiples of λ (that is, λ , $\lambda/2$, $\lambda/3$, etc.)

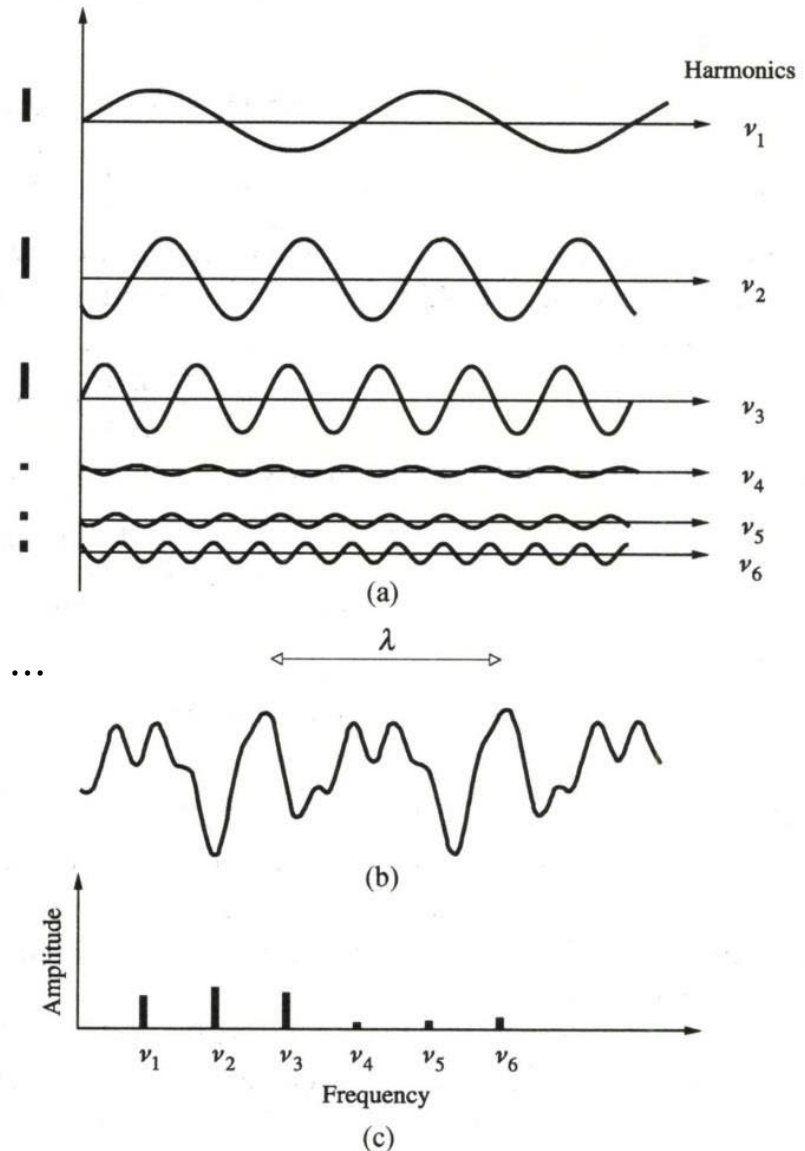
The Fourier-series representation:

$$f(x) = C_0 + C_1 \cos\left(\frac{2\pi}{\lambda}x + \epsilon_1\right) + C_2 \cos\left(\frac{2\pi}{\lambda/2}x + \epsilon_2\right) + \dots$$

where the C -value are constants

The profile $f(x)$ may correspond to a traveling wave $f(x - vt)$.

The synthesized function comprises an infinite number of terms, the series will presumably be identical to $f(x)$



The Fourier-series representation:

$$f(x) = C_0 + C_1 \cos\left(\frac{2\pi}{\lambda}x + \epsilon_1\right) + C_2 \cos\left(\frac{2\pi}{\lambda/2}x + \epsilon_2\right) + \dots$$

$$C_m \cos(mkx + \epsilon_m) = A_m \cos mkx + B_m \sin mkx$$

where $k = \frac{2\pi}{\lambda}$, $A_m = C_m \cos \epsilon_m$, $B_m = -C_m \sin \epsilon_m$

$$f(x) = \frac{A_0}{2} + \sum_{m=1}^{\infty} A_m \cos mkx + \sum_{m=1}^{\infty} B_m \sin mkx$$

The process of determining the coefficients A_0 , A_m , and B_m for a specific periodic function $f(x)$ is referred to **Fourier Analysis**.

$$\int_0^\lambda \sin mkx \, dx = \int_0^\lambda \cos mkx \, dx = 0$$

$$\int_0^\lambda f(x) \, dx = \int_0^\lambda \frac{A_0}{2} \, dx = A_0 \frac{\lambda}{2}$$

$$A_0 = \frac{2}{\lambda} \int_0^\lambda f(x) \, dx$$

$$\int_0^\lambda \sin akx \cos bkx \, dx = 0$$

$$\int_0^\lambda \cos akx \cos bkx \, dx = \frac{\lambda}{2} \delta_{ab}$$

Kronecker delta

$$\int_0^\lambda \sin akx \sin bkx \, dx = \frac{\lambda}{2} \delta_{ab}$$

$$\int_0^\lambda f(x) \cos mkx \, dx = \int_0^\lambda A_m \cos^2 mkx \, dx = \frac{\lambda}{2} A_m$$

$$A_m = \frac{2}{\lambda} \int_0^\lambda f(x) \cos mkx \, dx$$

$$B_m = \frac{2}{\lambda} \int_0^\lambda f(x) \sin mkx \, dx$$

A periodic function $f(x)$ can be represented as a Fourier series

$$f(x) = \frac{A_0}{2} + \sum_{m=1}^{\infty} A_m \cos mkx + \sum_{m=1}^{\infty} B_m \sin mkx$$

Knowing the function $f(x)$, the coefficients A_m and B_m can be computed using the following formula

$$A_0 = \frac{2}{\lambda} \int_0^{\lambda} f(x) dx$$

$$A_m = \frac{2}{\lambda} \int_0^{\lambda} f(x) \cos mkx dx$$

$$B_m = \frac{2}{\lambda} \int_0^{\lambda} f(x) \sin mkx dx$$

For even function $f(x)$

$$f(-x) = f(x); B_m = 0 \text{ for all } m.$$

For odd function $f(x)$

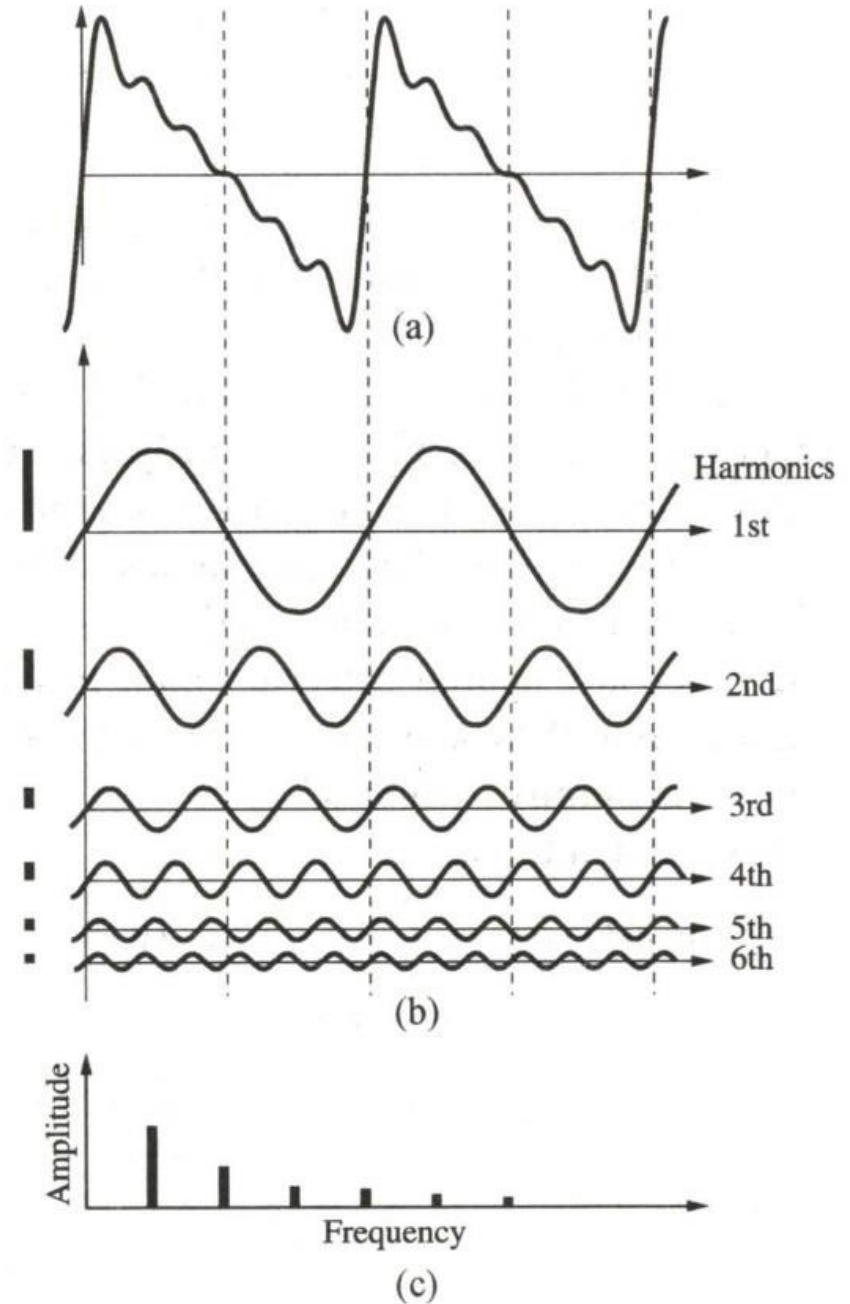
$$f(-x) = -f(x); A_m = 0 \text{ for all } m.$$

Example: the saw-tooth function

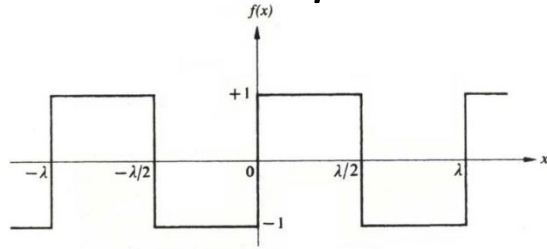
$f(-x) = -f(x)$; so $A_m = 0$ for all m .

The Fourier-series representation:

$$f(x) = \frac{A_0}{2} + \sum_{m=1}^{\infty} B_m \sin mkx$$



Example: compute the Fourier series that corresponds to a *square wave*.



$$f(x) = \begin{cases} +1 & \text{when } 0 < x < \frac{\lambda}{2} \\ -1 & \text{when } \frac{\lambda}{2} < x < \lambda \end{cases}$$

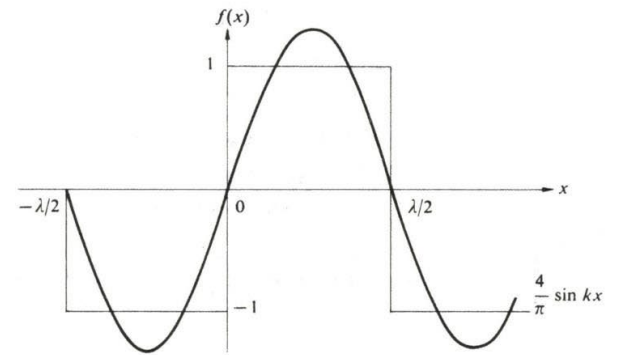
$$B_m = \frac{2}{\lambda} \int_0^{\lambda/2} (+1) \sin mkx \, dx + \frac{2}{\lambda} \int_{\lambda/2}^{\lambda} (-1) \sin mkx \, dx$$

$$B_m = \frac{1}{m\pi} [-\cos mkx]_0^{\lambda/2} + \frac{1}{m\pi} [\cos mkx]_{\lambda/2}^{\lambda}$$

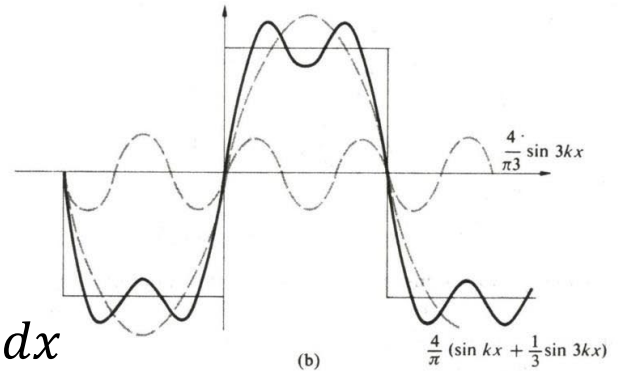
$$B_1 = \frac{4}{\pi}, \quad B_2 = 0, \quad B_3 = \frac{4}{3\pi}, \quad B_4 = 0, \quad B_5 = \frac{4}{5\pi}, \dots$$

The required series:

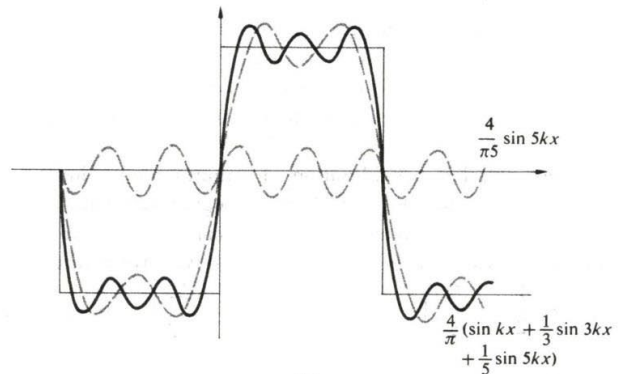
$$f(x) = \frac{4}{\pi} \left(\sin kx + \frac{1}{3} \sin 3kx + \frac{1}{5} \sin 5kx + \dots \right)$$



(a)



(b)



(c)